DATA 583

Advanced Predictive Modelling

Random vectors and their expectations

If $Z_1, Z_2, \ldots, Z_n$ is a collection of random variables, we can define a random vector Z as

$$Z^\top = [Z_1\ Z_2 \cdots Z_n] \tag{1}$$

The expected value of Z is defined as the vector of expected values of Z:

$$E[Z^\top] = [E[Z_1]\ E[Z_2] \cdots E[Z_n]]. \tag{2}$$

The variance-covariance matrix

A useful summary of the variability in a random vector is provided by the variance-covariance matrix.

It also provides a view of a special type of dependence, i.e. linear dependence among pairs of elements in the random vector.

$$\textbf{Var}(Z) = E[(Z - E[Z])(Z - E[Z])^{\top}]. \tag{3}$$

Linear combinations of random variables

- If a random vector is multiplied by a scalar constant, its expectation is also multiplied by that constant:

$$E[aZ] = aE[Z]. \tag{4}$$

- If A is an $m \times n$ matrix, then

$$E[AZ] = AE[Z]. \tag{5}$$

Note that the number of columns of A must match the number of elements in Z.
When the matrix A consists of only 1 row, AZ reduces to an inner product of A and Z. That is, if a is an n-vector, then

$$a^\top Z = \sum_{j=1}^{n} a_j Z_j. \tag{6}$$

The variance-covariance matrix of AZ

- **A key step here required the recollection from linear algebra that $(AB)^\top = B^\top A^\top$.**

- $$\textbf{Var}(AZ) = A\textbf{Var}(Z)A^\top. \tag{7}$$

$$\textbf{Var}\left(\sum_{j=1}^{n} a_j Z_j\right) = \sum_{j=1}^{n}\sum_{i=1}^{n} a_i a_j \textbf{Cov}(Z_i, Z_j) \tag{8}$$

- **The mathematical benefit of a constant variance**

$$\textbf{Var}(AZ) = A\sigma^2 I A^\top = \sigma^2 A A^\top. \tag{9}$$

- **Orthogonal transformations: $AA^\top = I$, where I is the identity matrix**

$$\textbf{Var}(AZ) = A\sigma^2 I A^\top = \sigma^2 A A^\top = \sigma I. \tag{10}$$